COMMERCIAL ITEM DESCRIPTION

VOLTMETER, AC

The General Services Administration has authorized the use of this Commercial Item Description.

Scope. This Commercial Item Description (CID), describes an AC voltmeter. This CID is meant as a minimum requirement for the AC voltmeter in which only those manufacturers that meet or surpass the following requirments are qualified per this CID.

Classsification. Equipment covered by this CID may be commercially available equipment modified to the extent necessary to meet the following description. The equipment shall be type III, class 5, style G and color R in accordance with MIL-T-28800.

Applicable documents.

Specifications and standards. The following specifications and standards form a part of this CID to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-T-28800	Test Equipment for use with Electrical and Electronic Equipment, General Specification for.

(Copies of specifications, standards, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

Other publications. The following document forms a part of this CID to the extent specified herein. Unless otherwise specified, the issue of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of

AMSC N/A FSC 6625

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documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI-C62.41-1980

(Application for copies of ANSI publications should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

Order of precedence. In the event of a conflict between the text of this CID and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this CID shall take precedence. Nothing in this CID, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

Salient characteristics. The equipment shall be capable of operation and be bid sample tested within the accuracies, limits, and specifications herein.

1. Safety. The equipment shall be UL (Underwriters Laboratories) listed and approved and/or shall comply with the type III safety requirements of MIL-T-28800.

a. Leakage current protection. Protection shall be provided from leakage current in excess of 3.5 milliamperes (mA) rms (AC and DC) from any accessible conductive parts of equipment (including control shafts with knobs broken or removed and recessed calibration or adjustment controls) to either pole of the power source for any position of the power switch(es).

b. Radio frequency interference (RFI) capacitors. Any capacitor used in RFI filters shall be rated for at least four(4) times the operating voltage.

c. Type III ventilation openings.

(1) Ventilation openings shall be designed to prevent accidental contact with live electrical circuits and mechanical moving parts.

(2) Top ventilation openings shall incorporate small mesh screening or similar design to prevent the expulsion of hot electronic particles due to failure of electronic devices such as capacitors.

2. Parts, materials and processes. Except for marking, the AC voltmeter shall comply with the type III parts, materials and process requirements of MIL-T-28800. Nonstandard parts approval requests shall be in accordance with MIL-T-28800. Front panel marking shall be in accordance with MIL-T-28800.

a. Restricted material. The equipment shall comply with the restricted material requirements of MIL-T-28800, including the requirement for mercury and radioactive materials, and shall contain no combination of materials which cause deterioration of any material contained in the equipment due to the effects of outgassing.

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3. Design and construction. Unless otherwise specified herein, the equipment shall comply with the type III design and construction requirements of MIL-T-28800. The equipment shall be of solid-state construction. Encapsulation and embedment (potting) of subassemblies may be used only for high voltage circuits where construction is susceptible to leakage or corona, indicators and associated controls shall be in close proximity to each other, all potentiometer controls shall have at least a 270 degree rotation range (4.71 radians).

a. The AC voltmeter shall come with one(1) blank space cover and one(1) extender board.

b. A package part number shall be assigned to the AC voltmeter as a whole system. This system part number shall be marked on the mainframe identification plate and plug-in(s).

c. All module interface shall be done in the back of the mainframe.

d. Signal input and output connectors. Unless otherwise specified, all signal input and output connectors shall be type BNC female.

e. Internal cooling. The equipment shall be capable of operating continuously for twenty four(24) hours at 55°C. If forced air is used, it shall be as specified in MIL-T-28800.

(1) Forced air cooling motor. DC brushless motors shall be utilized for the forced air cooling motor in the equipment.

(2) Air filter. Equipment utilizing a cooling scheme which exhausts air from the equipment through the cooling fan shall not require an externally accessible air filter.

(3) Thermal switch. A thermal switch shall be provided which will interrupt circuit power to the AC voltmeter when the internal temperature rises to the degree of possible damage to the electronic components. A minimum of 10°C hysteresis drop from this tripping temperature must occur before circuit power is again applied.

4. Electrical power sources. The equipment shall operate from nominal commercial, military, and shipboard power sources of 100, 110, 120, 200, 220 and 240 volts ac (±10%) at line frequencies of 50 Hz to 400 Hz.

a. Electrical power connectors. The equipment shall operate with a ground at the primary source on either side of the single phase input.

b. Power consumption. Maximum power consumption shall not exceed 6 watts.

c. Fuses. Fuse(s) shall be provided, on the exterior rear panel for replacement without entering the enclosure. The fuse(s) shall automatically disconnect the equipment from the power source in the event of excessive current requirements.

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d. Surge and transient voltage protection. The AC voltmeter shall contain surge and transient voltage protection internal of the enclosure. The transient voltage protection circuit shall be so designed as to prevent electronic device damage within the system per type III solid state tolerance bands of MIL-T-28800, the surge voltage protection circuit shall conform to ANSI/IEEE standard C62.41-1980. The surge protection circuit shall drain excess power due to surges directly to the ground conductor for personnel protection. Normal operation of the AC voltmeter may be inhibited and false operational signals may be generated under input voltage transients greater than ±20 percent and frequency transient conditions greater than ±5 percent from within the nominal type III steady-state tolerance bands of MIL-T-28800, damage (other than fuse failure) to the AC voltmeter shall not occur during voltage transient conditions greater than ±20 percent and frequency transient conditions greater than ±5 percent within the maximum limits (30 percent voltage transients and 10 percent frequency transients) specified in MIL-T-28800. Normal operation of the AC voltmeter shall be automatically resumed when the transient has passed and any blown fuses are replaced. Blown fuses shall be externally accessible. Testing shall be in accordance with MIL-T-28800.

5. Dimensions and weight. The size and weight of the equipment shall be consistent with current commercial capabilities, refer to paragraphs 5.a and 5.b of this CID.

a. Equipment dimensions. The nominal dimensions for the AC voltmeter should be approximately 5 ¼ inches height x 8 ½ inches wide x 12 inches deep.

b. Equipment weight. Equipment weight including all accessories excluding manuals, shall not exceed 6 lbs.

6. Enclosure requirements. Unless otherwise specified herein, the equipment shall comply with the style F enclosure requirements of MIL-T-28800.

a. Rackmount. Rackmount capability shall be available as an option for the equipment specified herein.

b. Tilt bail handle. A tilt-bail handle shall be provided in lieu of the hinged metal handles in accordance with MIL-T-28800. Exception to this paragraph may be acceptable as this is not a prime consideration of the equipment.

c. Reusable pouch or container. An attached accessory pouch may be used for storage of all supplied accessories in lieu of front-cover storage.

7. Marking and identification.

a. Marking. The visible surface adjacent to panel facilities such as variable controls, indicators, jacks, connectors and fuse holder(s), shall be marked with a suitable word, phrase, or abbreviation thereof, indicating the purpose or use of the part.

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(1) Panel marking. The front panel marking shall comply with the type III parts, materials and processes panel marking requirements of MIL-T-28800; however, marking identification processes other than the type III requirements specified in MIL-T-28800 may be used, provided the marking and identification processes are capable of passing the front panel marking test when tested in accordance with this CID.

b. Items to be marked.

(1) Volume and weight (including accessories) shall be marked on three external surfaces of the enclosure.

(2) The type designation of the AN nomenclature of the AC voltmeter shall be stenciled on the exterior or main frame on at least three adjacent surfaces, on the transient case on two opposite sides. Marking shall be in block letters approximately 2 inches in height.

8. Environmental requirements.

a. Temperature requirements. Unless otherwise specified herein, the equipment shall meet its specified performance characteristics and other requirements when subjected to the class 2 temperature requirements specified by MIL-T-28800 and as listed below.

(1) Non-operating temperature. The equipment shall meet its performance and accuracy requirements in an non-operating environment of -40°C to +85°C.

(2) Operating temperature. The equipment shall meet its performance and accuracy requirements in an operating environment of 0°C to 55°C.

b. Humidity requirements. The equipment shall meet its required performance and accuracy requirements under humidity requirements for class 3 equipment as specified by MIL-T-28800.

c. Altitude requirements. The equipment shall meet its required performance and accuracy requirements under the altitude requirements for class 2 equipment as specified by MIL-T-28800.

d. Vibration requirements. The equipment shall meet the class 2 vibration requirements as specified by MIL-T-28800.

e. Mechanical shock requirments.

(1) Shock, pulse. The equipment shall meet the class 7 shock, pulse requirement as specified by MIL-T-28800.

(2) Transit drop. The equipment shall meet transit drop requirement as specified by MIL-T-28800.

(3) Bench handling. The equipment shall meet the requirements as specified by MIL-T-28800.

(4) Shock, high impact. The equipment shall meet the requirements as specified by MIL-T-28800.

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f. Electromagnetic interference (EMI). The equipment shall comply with the following EMI requirements specified in MIL-STD-461B:

(1) CE01: Narrow band requirement from 10kHz to 20kHz only.

(2) CS03: Full limits.

(3) CS01 Full limits.

(4) CS02: Full limits.

(5) CS06: Full limits.

(6) RE01: Relaxed 10dB at the fundamental, 3rd harmonic, and 5th harmonic of the source power frequency with the measurement probe located 12 inches from device under test.

(7) RE02: Full limits to 10 times the highest used or intentionally generated frequency, or 1GHz, whichever is greater.

(8) RS01: Full limits.

(9) RS03: Full limits to 1GHz.

g. Magnetic environment. The equipment shall comply with the magnetic environment requirements of MIL-T-28800.

9. Reliability. Reliability verification testing, shall be in accordance with the type III, class 3 environmental conditions of MIL-T-28800.

a. Mean time between failures (MTBF). The equipment mean time between failures (MTBF) shall be 2,000 hours. The lower test MTBF shall be 2,000 and the upper test shall be MTBF 4,000.

b. Pre-conditioning requirement. Each deliverable equipment shall be subjected to the environmental screening effects of burn-in and thermal cycling, with thermal cycling being also accomplished in the sample testing phase.

(1) Burn-in. Burn-in shall expose the equipment to extended periods of elevated temperatures, (43°C for a minimum of 96 hours). Each equipment which fails during burn-in shall be repaired and fully retested, until failure free. Failures which occur during the burn-in test shall be noted and reported but shall not count toward the establishment of MTBF. Prior to and following burn-in, a satisfactory operation test (SOT) shall be performed. A satisfactory operation check (SOC) shall be performed conducted at selected times during the burn-in test.

(2) Thermal cycling. Thermal cycling shall expose the equipment to hot and cold cycles for class 3 equipment per MIL-T-28800. Each equipment which fails during thermal cycling shall be repaired and full cycling test shall be noted and reported but shall not count toward the cycling, a satisfactory operation check (SOC) shall be conducted during the last two upper and lower temperature limits.

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10. Calibration and maintenance adjustments. The design of the system shall provide for readily accessible calibration adjustments and maintenance adjustments. These adjustments shall be provided by variable value components which are adjustable by the use of simple means. The calibration by substitution of selected components or parts is unacceptable unless specifically approved. The calibration adjustments, wherever possible, shall be accessible without removal of the instrumentcase or modules.

a. Maintainability. The equipment shall meet the type III maintainability requirements specified by MIL-T-28800.

11. Interchangeability. The equipment shall meet the type III interchangeability requirements specified by MIL-T-28800.

12. Performance characteristics.

a. Input protection. The maximum input, without damage, shall be at least 800V peak, not to exceed a volt-hertz product of 1 x 10 on any range.

b. Input impedance. 2 or 10 megohm shunted by a nominal 25 picofarads.

c. Voltage ranges. 300 uV millivolt to 300 volts true RMS full scale, in 6 or more ranges at frequencies from 10 Hz to 10 MHz.

d. Resolution. The resolution shall be at least 0.05% of range.

e. Voltage accuracy. The ac voltage accuracies shall be within those specified in Table I.

TABLE I. Accuracies (21°C - 25°C).

Function	Range	Accuracies ± (of full scale)
AC volts		
10Hz-20Hz	all	5%
20Hz-50Hz	all	1%
	2mV	3%
	20mV	2%
50Hz-200kHz	---	0.5%
	20mV	2%
	2mV	3%
200kHz-1MHz	---	0.7%
	20mV	2%
	2mV	3%
1MHz-20MHz	---	3%
10MHz-20MHz	---	10%

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f. Decibel range. -72 to 52 dBm (0 dBm = mW of 600 ohm).

g. Decibel resolution. 0.01 dB.

h. Decibel accuracy. ± 0.45 dB 10 Hz and 1 MHz to 20 MHz. These measurements are above 10% of the range and the temperature is not above 10% of the range and the temperature is not above 38°C or below 8°C.

i. Crest factor. (Ratio of peak-to-rms amplitude of input signal) at least 4:1 at full scale, 18.1 at 1/3 full scale.

j. Max common mode voltage. 400mV rms, 600mV peak.

k. AC common mode rejection. 60 dB at 50 Hz and 60 Hz with 100 ohm unbalance.

l. Output noise. Less than 1 mV rms.

13. Calibration interval. After calibration, the equipment shall meet each performance requirement within the tolerances specified for a period of at least 9 months, based on an operating time of 1500 hours.

Certification. The contractor shall certify that the product offered meets the salient characteristics of this description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices, and is the same product offered for sale in the commercial marketplace. Small businesses that cannot offer products with established commercial market acceptability, may compete by submission of bid samples. (The Government reserves the right to require proof of such conformance before first delivery and thereafter as may be otherwise provided for under the provisions of the contract).

Packaging, packing and marking. The items shall be packaged in accordance with normal commercial practice and packed to assure acceptance by common carrier and provide product protection against loss and damage during multiple shipments, handling and storage. The shipping container shall be in compliance with national motor freight classification and uniform freight classification. Shipping container shall be marked as specified in the contract or order.

Known acceptable products. The following product(s) are known to have previously met the requirements of this CID and need not resubmit bid samples or certification. All other manufacturers must meet the requirements of this CID and submit to bid testing and certification as required.

Manufacturer	Model	NSN
Ballantine	3046A	6625-01-010-3514
Boonton	93AD	6625-01-050-8696
Fluke	8920A	6625-01-081-2638
	8921A	6625-01-203-1314

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